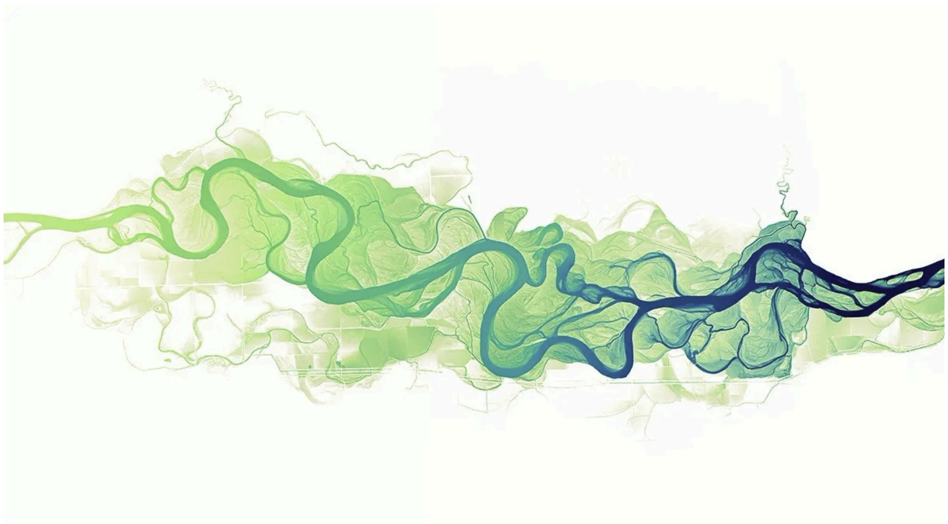
ZURICH UNIVERSITY OF APPLIED SCIENCES
DEPARTEMENT LIFE SCIENCES AND FACILITY MANAGEMENT

Automated Detection and Morphological Classification of Mast Cells in Bone Marrow Aspirates Using Deep Learning



Bachelor Thesis

by
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Abstract

English Version

German Version

Acknowledgement

I would like to express my sincere gratitude to Dr. Stefan Glüge for setting up the project for this thesis and for his valuable support and guidance throughout the writing and compilation process of this thesis. My appreciation also goes to Robin Kosche for showing me around the USZ and introducing me to the subject.

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List of Abbreviations

- **AI**: Artificial Intelligence
- **AP**: Average Precision
- **CNN**: Convolutional Neural Network
- **COCO**: Common Objects in Context
- **CV**: Cross Validation
- **CVAT**: Computer Vision Annotation Tool
- **DoE**: Design of Experiment
- **FN**: False Negative
- **FP**: False Positive
- **IoU**: Intersection over Union
- **LOPO**: Leave-One-Patient-Out
- **MC**: Mast Cells
- **mAP**: Mean Average Precision
- **NMS**: Non-Maximum Suppression
- **RF-DETR**: Roboflow Detection Transformer
- **SM**: Systemic Mastocytosis
- **TN**: True Negative
- **TP**: True Positive
- **USZ**: University Hospital Zurich (Universitätsspital Zürich)
- **WHO**: World Health Organization
- **YOLO**: You Only Look Once

1. Introduction

2. Methodology

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General Data Information

The dataset for this study consists of digitized bone marrow aspirate smear images provided by the University Hospital Zurich (USZ). All samples were collected from patients with confirmed or suspected systemic mastocytosis. Image tiles were extracted from whole-slide scans at a fixed resolution of 512x512 pixels, each displaying a sub-section of the whole smear. Two morphological classes were labeled for this study: Atypisch (atypical mast cells) and Normal (normal mast cells), corresponding directly to the cell types relevant to the WHO minor diagnostic criterion for SM.

The corpus grew across four acquisition batches spanning 53 patients. The initial dataset covered a single patient (P1) and served as the pilot for the annotation and training pipeline. A second batch extended the corpus to 15 patients (P1-P15), contributing 1,564 annotated images with 1,288 Atypisch and 391 Normal instances, reflecting a raw class ratio of approximately 3.3:1. A third batch added patients P16 through P53, bringing the full corpus to 53 patients (1,857 annotated images, 1,476 Atypisch, 511 Normal, raw ratio 2.9:1). Because the training data derives almost entirely from confirmed SM patients, Atypisch is the dominant class throughout. At the P1-P8 scale the raw ratio was 14.4:1; by P1-P15 this had dropped to 3.3:1 due to the inclusion of patients P6, P8, and P9, which are rich in Normal annotations.

Data Acquisition and Annotation

The annotation pipeline was carried out by Robin Kosche at USZ using CVAT (Computer Vision Annotation Tool). Rather than annotating all 53 patients manually from scratch, an iterative model-assisted annotation strategy was used to make the labeling process tractable at the scale of tens of thousands of image tiles.

The first patient (P1) was annotated entirely by hand. Bounding boxes were drawn around each visible mast cell. This initial patient did only had Atypisch cells. This dataset was used to train the initial domain-specific detection model, DL_Modell_FV.pt, a YOLO11n checkpoint trained on P1 data.

For patients P2 through P15, CVAT's model inference integration was used with DL_Modell_FV.pt as the backend. The model generated candidate bounding boxes for each image tile, which the annotator then reviewed, accepted, corrected, or rejected. Starting from this batch, also normal cells were annotated. Tiles where the model produced detections on regions that contained no mast cell were recorded separately in per-patient Excel files. These confirmed false positives were later incorporated into training as a curated set of hard-negative examples.

For patients P16 through P53, the same workflow was repeated using a second-generation model (DL_Modell_FV_v2.pt), trained on the P1-P15 corpus, as the inference backend. It is worth noting that a subset of patients in this batch contained no annotatable mast cells at all: the model produced candidate detections, but all were rejected by the annotator as false positives. These patients contributed no positive labels to the corpus but supplied a large number of confirmed hard negatives. The annotations from the full P1-P53 corpus were then used to train a third-generation model, DL_Modell_FV_v3.pt, completing the bootstrapped annotation cycle at the current project scope.

This iterative approach, where each annotation round produces a labeled dataset used to train a better model for the next round, allowed the corpus to grow from one patient to 53 without requiring fully manual annotation of all 53 cases. The confirmed false positives collected at each round provided a particularly valuable training signal: these are the specific tiles that the model of that generation found difficult to reject, making them targeted hard negatives for the next training cycle.

This bootstrapped annotation strategy is not documented for mast cell detection in the existing literature. The systematic review by Ghete et al. [1], covering AI-based bone marrow cell classification from 2019 to 2024, found that the majority of published studies applied their models to pre-existing datasets rather than developing annotation pipelines alongside the model. In this project, annotation and model development were tightly coupled by design: the clinical partner's annotation capacity was directly extended by each generation of model, and the confirmed false positives produced during annotation became training signal for the next generation. This coupling is what made a 53-patient corpus feasible within the scope of a single thesis.

The final corpus assembled across all three annotation rounds is summarized in Table 1.

Table 1: Full corpus summary across P1-P53.

Category	Count
Total annotated images	1,857
Atypisch cell labels	1,476
Normal cell labels	511
Raw class ratio (Atyp:Norm)	2.9 : 1
Confirmed FP/Background images	4,940

Data Preprocessing

Two distinct preprocessing approaches were applied across the experimental timeline, with a significant revision midway through the project.

Pre-applied augmentation (Phase 1 and early Phase 2). For the initial experiments, a preprocessing pipeline was implemented using the albumentations library. Each annotated image was transformed once before training with random flips, brightness and contrast adjustments, and affine rotations of up to 15 degrees. The augmented copies were saved to disk and used as fixed training input. This approach proved ineffective: because transforms were applied once and stored, every training epoch saw the same augmented copies. The augmented images extended the dataset by a fixed factor but provided no additional regularization benefit after the first epoch, which is the only epoch where the transforms are novel to the model. This limitation was identified during Phase 2 and motivated the switch to on-the-fly augmentation.

On-the-fly augmentation (Phase 3 onward). Starting from Phase 3, augmentation was moved into the YOLO training loop by enabling augment=True in the model.train() call. This applies a fresh random transform to each image at every epoch, meaning oversampled copies in the training file list present different pixel content to the model across epochs. Two augmentation parameters required explicit configuration for this domain. The mosaic augmentation was disabled (mosaic=0.0): YOLO's mosaic mode composites four images into a single training sample, which is appropriate for multi-object scene datasets like COCO but is counterproductive for 512x512 bone marrow tiles, where the spatial relationship between a candidate cell and its immediate local context is part of the morphological signal the model needs to learn. Vertical flip probability was set to 0.5 (the YOLO default is 0), consistent with the biological reality that bone marrow mast cells have no canonical vertical orientation and that the model should not learn a vertical bias from the training data.

The rotation augmentation parameter (degrees) was treated as a tunable hyperparameter during Phase 3. A systematic tinkering matrix identified degrees=5 as optimal at the Phase 3 corpus scale, providing approximately 4-5 percentage points of improvement in Normal-class recall over degrees=0. The effect is biologically justified: mast cells present in any orientation in aspirate smears, and moderate rotation augmentation regularizes the model without distorting morphological cues that distinguish Atypisch from Normal.

Model Selection and Architecture

YOLO11 (Ultralytics) was selected as the object detection framework. The architecture performs detection and classification in a single forward pass, making it computationally efficient for integration into clinical workflows where inference throughput matters. The Ultralytics Python API provides a consistent interface for training, fine-tuning, evaluation, and inference, and supports checkpoint initialization from custom pretrained weights, which was central to the transfer learning strategy described below. Comparable studies applying YOLO-family models to cell detection in hematological microscopy report mAP50 values in the range of 77-94% on well-represented cell types [2] [3], placing the architecture in the performance range expected for this domain.

Among the available YOLO11 model sizes, the nano variant (yolo11n, 2.6M parameters) was selected. The selection was validated experimentally: a test using the large variant (yolo11l, 43M parameters) on the P1-P8 corpus under identical training conditions produced R_Normal=0.695 compared to R_Normal=0.840 for yolo11n. The larger model's classification head, which is randomly re-initialized when training with a different number of output classes than the pretrained checkpoint, has proportionally more free parameters to adapt from scratch. With only 170-220 training images per LOPO (Leave-One-Patient-Out) fold, these parameters tend to overfit to the majority class rather than learn generalizable features for the rare Normal class. The corpus size at this stage was insufficient for a 43M-parameter model; yolo11n was confirmed as the appropriate architecture for the available data.

Throughout the project, all training runs started from the generic COCO-pretrained yolo11n.pt checkpoint. The single exception was a set of experiments during Phase 2 in which DL_Modell_FV.pt, a YOLO11n model previously trained on the P1 single-patient dataset (nc=1, one class), was used as the starting point for continued training on the two-class P2 dataset. The reasoning was that domain-specific pretraining might give the model a better starting point for bone marrow aspirate images compared to COCO features. In practice, this approach introduced a structural complication: because DL_Modell_FV.pt was trained with a single output class and the P2 training required two classes (Atypisch and Normal), the Detect head's classification branch (cv3, the final 1x1 convolution layer) had to be discarded and randomly re-initialized. The bounding-box regression branch (cv2) transferred cleanly, but the randomly initialized cv3 layer produced noisy classification gradients in early training epochs. At the default learning rate of lr0=0.01, these gradients were large enough to partially overwrite the pretrained neck features, causing systematic underperformance rather than the expected improvement. The full account of this failure and its resolution is described in the training strategy section. After Phase 2, the project returned to yolo11n.pt as the standard base for all subsequent training.

Model Training and Finetuning Strategy

The training strategy developed across five phases, each responding to specific failures and insights from the previous phase. The following account describes the major decision points rather than every individual run; a complete run table is provided in the Results section. The phase structure corresponds to successive expansions of the training corpus and methodology: Phase 1 covered a single pilot patient, Phase 2 moved to a two-class annotated dataset and explored data composition, Phase 3 introduced multi-patient LOPO cross-validation and on-the-fly augmentation on an 8-patient corpus, Phase 4 expanded to 15 patients, and Phase 5 targets the full 53-patient corpus. Individual experiment runs within each phase are identified by a label combining the phase number and a letter (e.g., P2-C, P3-A), which matches the experiment log maintained throughout the project.

Phase 1: Single-patient proof of concept and hyperparameter search.

Phase 1 covered a sequence of experiments on the P1 dataset (428 images, single patient, Atypisch class only) using 5-fold stratified cross-validation and the pre-applied albumentations augmentation pipeline described above. The dataset consisted of annotated positive tiles alongside hand-curated negative tiles (confirmed empty regions), all at 512x512 pixels.

The initial runs used yolo11n.pt (COCO pretrained). To test whether a larger model would learn better features for this domain, yolo11l (43M parameters) was then substituted and run in three separate configurations, including extended epoch budgets (200 epochs) and added augmentation. In each case, the large model did not outperform yolo11n on this single-patient dataset, and the project returned to yolo11n for all further Phase 1 work. This finding prefigured the same result later confirmed empirically at larger scale in Phase 3 (P3-E), where yolo11l again underperformed yolo11n on a multi-patient corpus.

With yolo11n established as the appropriate architecture, hyperparameter tuning was run using YOLO's built-in model.tune() function across three increasingly intensive rounds: 30 iterations x 50 epochs, 300 iterations x 50 epochs, and finally 300 iterations x 100 epochs. The output of the final tuning round was a best_hyperparameters.yaml file with optimized values including lr0=0.0062, momentum=0.791, mosaic=0.916, and degrees=0.0. A training run applying these tuned hyperparameters completed Phase 1. The per-fold results are reported in the Results section.

The best_hyperparameters.yaml produced by the Phase 1 tuner is significant beyond Phase 1: it was later applied to fine-tuning on P2 data (P2-E) and caused catastrophic forgetting, collapsing performance across all five folds. The failure was later attributed to a recipe mismatch: the tuner had optimised on a from-scratch yolo11n training run, and its output (particularly the high mosaic value and degrees=0.0) was calibrated for scratch training from COCO, not for fine-tuning in this clinical domain. The Phase 1 yaml was permanently retired from the fine-tuning pipeline as a result.

Following the 5-fold training, an inference run was conducted on the full first-patient slide image set, approximately 20,000 tiles, excluding all training images by filename. The resulting confidence distribution provided an early look at the model's behavior at deployment scale and informed the direction of subsequent annotation and training work.

Phase 2: The domain-specific checkpoint experiment and the learning rate problem.

Phase 2 began with two runs using the standard yolo11n.pt base on the annotated P2 dataset (P2-A, P2-B), but both were corrupted by a concurrent SLURM job collision described below.

The third run (P2-C) introduced DL_Modell_FV.pt as the starting point, testing whether the P1-trained domain-specific model would give the detector a better foundation than COCO-pretrained weights. The result was a characteristic failure pattern: high precision combined with very low recall, the signature of a model that makes few predictions and misses most cells. The cause was the interaction between the randomly re-initialized cv3 head and the default learning rate of lr0=0.01: the noisy gradients from the freshly initialized classification branch were large enough to partially overwrite the pretrained neck features during early epochs, leaving the model worse than yolo11n.pt would have been from scratch.

Reducing the learning rate to lr0=0.001 recovered the performance lost to this overwriting effect. Freezing the first ten backbone layers (freeze=10) was introduced alongside the learning rate change to prevent the small dataset from further disrupting features that COCO pretraining had established. This combination (lr0=0.001, freeze=10) defined the reference configuration for the remainder of Phase 2. However, the conclusion drawn from the full Phase 2 experiment set was that using DL_Modell_FV.pt as a starting point added complexity without a measurable benefit over yolo11n.pt, and all subsequent phases returned to the standard COCO-pretrained base.

Before expanding to the full P1-P15 corpus that would produce the second model version, a 3x4 factorial design-of-experiment (DoE) grid was run to identify the best data composition strategy at Phase 2 scale. The goal was practical: the hyperparameter and data configuration choices made here would carry forward into the P1-P15 training, and a better second model version meant fewer manual corrections for the annotation partner at USZ when working through patients P16-P53. The grid varied confirmed FP negative oversampling (1x, 2x, 3x) and random background tile ratio (0, 1, 2, 3) across twelve configurations. The mAP50 spread across all twelve DoE configurations was smaller than the cross-fold standard deviation within any single run. Data composition manipulations at single-patient scale could not produce statistically interpretable performance differences; the limiting factor was data scarcity rather than class balance. Background tiles uniformly hurt performance at every FP oversampling level, and this finding was consistent enough to permanently set the random background ratio to zero in all subsequent runs. The DoE finding reframed the problem: improving recall required more patients and better augmentation, not a different ratio of existing data categories. This result has broader relevance for rare-event medical imaging tasks where the class imbalance is not accidental but structural: when a dataset is collected from disease-positive patients, the minority class is underrepresented because the disease itself is rare in that morphological form, not because of sampling bias. In that regime, reweighting or resampling the existing data cannot substitute for collecting more biologically diverse examples. The systematic nature of the DoE grid makes this conclusion more than a project observation; it is a controlled negative result that points practitioners toward corpus expansion as the primary lever for rare-class recall improvement.

Phase 2 also exposed an infrastructure failure specific to the HPC environment. Submitting two SLURM jobs concurrently caused both jobs to share the same temporary workspace directories, because directory paths were derived from fold indices without job-level namespacing. Concurrent writes to shared directories corrupted three of five folds in each job, detectable by fold results that were bit-for-bit identical between the two affected runs. The fix applied $SLURM_JOB_ID namespacing to all temporary directories and derived the output project directory name from the job name set at sbatch submission time, making concurrent jobs structurally independent.

Phase 3: Multi-patient LOPO and on-the-fly augmentation.

Phase 3 restructured the training and evaluation design around three simultaneous changes: expansion to 8 patients, adoption of Leave-One-Patient-Out cross-validation, and the switch to on-the-fly augmentation.

The LOPO design was motivated by a systematic bias in the Phase 1 and Phase 2 stratified k-fold protocol. When images from the same patient appear in both training and validation splits, the model can exploit patient-specific staining characteristics and morphological style to improve validation metrics without learning features that generalize to new patients. LOPO eliminates this by holding out all images from a single patient per fold, making each fold an evaluation of genuine cross-patient generalization.

The combined effect of these three changes was substantial. The Phase 3 baseline run (P3-A: 8 patients, yolo11n.pt, augment=True, mosaic=0.0, cls=1.0) achieved the strongest results seen to that point in the project, with cross-fold standard deviation markedly lower than the Phase 2 DoE despite the harder evaluation protocol: the more diverse training corpus more than compensated for the stricter holdout design. Full results are reported in the Results section.

With the same goal of finding the best possible training recipe before the P1-P15 training that would produce the second model version, a tinkering matrix of 14 parameter combinations explored rotation degrees (0, 5, 10), distribution focal loss weight dfl (1.5, 2.0), backbone freeze depth (0, 10), and label smoothing (0.0, 0.1). The production recipe confirmed here would be directly applied to the P1-P15 training: a higher-quality second model version meant the annotation partner at USZ would need fewer manual corrections when processing patients P16-P53. Rotation degrees was the dominant factor: degrees=5 consistently improved R_Normal over degrees=0 across all freeze and dfl combinations. Mast cells present in all orientations in aspirate smears, and moderate rotation augmentation directly targets this. The dfl parameter showed a consistent directional effect: dfl=2.0 regressed R_Normal relative to dfl=1.5 across all other parameter combinations, suggesting that increasing localisation loss weight tilts model capacity away from the classification discrimination needed for the rare Normal class. Freeze depth confirmed the Phase 2 finding: freeze=10 outperformed freeze=0 in every recall-relevant comparison. Label smoothing proved inert in this Ultralytics version, likely because the implementation does not propagate the parameter through the classification loss path.

The production recipe confirmed by the tinkering matrix was: degrees=5, dfl=1.5, freeze=10, lr0=0.001, cls=1.0, augment=True, mosaic=0.0, flipud=0.5.

An automated hyperparameter search using YOLO's built-in model.tune() was run on the production recipe (30 iterations, 50 epochs, patient P4 as holdout, mAP50-95 as fitness). The tuner converged on degrees=0.0 and dfl=2.47, both directions that the tinkering matrix had identified as harmful to R_Normal. The explanation is structural: a single-fold tuner optimizing mAP50-95 on a P4 holdout (an Atypisch-dominant patient) maximizes Atypisch-class performance at the expense of Normal-class generalization. Single-fold automated tuning is blind to cross-patient per-class recall balance. The production recipe from the tinkering matrix was retained.

A model-size comparison test (P3-E) substituted yolo11l (43M parameters) for yolo11n (2.6M parameters) under identical conditions on the P1-P8 corpus. R_Normal dropped to the lowest value measured across the entire project. The larger model's randomly re-initialized classification head has more free parameters, and at approximately 170-220 training images per fold the model memorized Atypisch patterns rather than learning generalizable Normal features. This result confirmed yolo11n as the appropriate model size for the available corpus.

The production recipe confirmed by Phase 3 was carried forward and applied to train on the full P1-P15 corpus, producing the second model version (DL_Modell_FV_v2.pt) that the annotation partner at USZ used for patients P16-P53.

Phase 4: Corpus expansion to 15 patients.

Phase 4 expanded the training corpus to 15 patients (1,564 images, 1,288 Atypisch, 391 Normal, 2.464 Background images incl. FP; raw ratio 3.3:1, effective ratio 1.5:1 after automatic per-patient oversampling). The most significant addition was patient P9, which contributed 195 Normal annotations, more than doubling the total Normal count relative to the P1-P8 corpus. A 15-fold LOPO protocol was applied.

The first Phase 4 run (P4-A) was configured with degrees=0 due to an omitted configuration variable; the second run (P4-B) applied the full production recipe including degrees=5. The delta between the two runs was within the cross-fold standard deviation on all metrics, indicating that the benefit of degrees=5 observed at P1-P8 scale either diminishes with a more diverse 15-patient corpus or is within measurement noise at this scale. A freeze ablation run (P4-C, freeze=0) produced results essentially equivalent to P4-B across all metrics, confirming that backbone freezing remains appropriate and the P3-B recipe transfers to the larger corpus without re-tuning. Full per-fold results for P4-A, P4-B, and P4-C are reported in the Results section.

Phase 5: Full corpus (P1-P53). [Results pending]

The Phase 5 training run (P5-A) was submitted to the ZHAW HPC cluster during the final stages of this thesis. The run uses the confirmed production recipe (degrees=5, dfl=1.5, freeze=10, lr0=0.001, cls=1.0), fully automatic per-patient oversampling targeting a 2:1 Atypisch:Normal ratio, and a LOPO design across all annotated patients after filtering out patients with no mast cell labels. Results were not available at the time of writing and will be incorporated in the final version of this thesis.

Model Testing

Performance estimates in this thesis are reported under two different evaluation protocols depending on the phase. Phase 1 and Phase 2 used stratified k-fold cross-validation (5 folds), where images were split randomly across folds with class stratification. From Phase 3 onward, the evaluation protocol was changed to LOPO, where all images from one patient are withheld from training entirely and used exclusively for evaluation. The switch was motivated by a systematic bias identified in the Phase 1 and Phase 2 results: stratified k-fold splits on small medical imaging datasets allow patient-level texture and staining style to leak from training into validation, artificially improving validation metrics without reflecting true cross-patient generalization. LOPO results and stratified k-fold results are therefore not directly comparable and are reported separately in the Results section.

Choosing LOPO over stratified k-fold directly addresses a gap identified in the broader literature. Ghete et al. [1] found that performance metrics across published bone marrow cell classification studies are not directly comparable in part because most studies evaluate on subsets of their own data, with few reporting results on patient-held-out splits and fewer still validating externally. LOPO is the strongest within-dataset approximation of external validation available without access to a second institution's data: each fold evaluates the model on a patient it has never seen in any form, under the same conditions a deployed system would face in routine clinical use. The cross-fold standard deviation reported in this thesis therefore reflects genuine inter-patient biological variability rather than statistical noise from random sampling, making the performance estimates more conservative and more informative than single-split or random k-fold results typically reported in the field.

Each fold produced a training split, a validation split used only for early stopping during training, and a test split consisting of the held-out patient's images. The metrics reported throughout this thesis refer to the test split. Across folds, the mean and standard deviation of each metric were computed. The cross-fold standard deviation reflects genuine inter-patient biological variability; patients with very small annotated sets (for example, P13 with 7 images) produce test metrics with extreme sensitivity, where a single missed cell changes recall by more than 14 percentage points.

One measurement limitation was identified in the per-class recall extraction. The implementation indexed Ultralytics' per-class result arrays by position rather than by class label index. When one class was absent from both ground truth and predictions in a given fold, Ultralytics returned a shorter result array and positional indexing misassigned recall values. This affected primarily the P13 holdout fold in every run, where the reported R_Normal was NaN while the true Normal recall appeared in the R_Atypisch position. Folds affected by this issue are excluded from per-class mean calculations throughout this thesis, and the directions of the per-class trends are verified against folds where the per-class counts are large enough to confirm correct assignment.

Evaluation Metrics

Model performance was assessed using standard metrics from object detection. Each detection outcome is first classified into one of four categories based on whether the model's predicted bounding box overlaps sufficiently with a ground-truth annotation (at a minimum Intersection over Union of 0.50):

- **True Positive (TP):** the model correctly detected a mast cell and the predicted box sufficiently overlapped the ground-truth annotation.
- **True Negative (TN):** the model correctly produced no detection in a tile containing no annotated mast cell.
- **False Positive (FP):** the model predicted a mast cell where the ground truth showed none (a false alarm).
- **False Negative (FN):** the model failed to detect a mast cell that was present in the ground truth (a missed cell).

From these, the following performance indicators were computed [PA2_FV]:

Precision measures the proportion of predicted detections that were correct:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall measures the proportion of ground-truth mast cells that the model successfully detected:

$$\text{Recall} = \frac{TP}{TP + FN}$$

mAP50 (Mean Average Precision at IoU 0.50) is the mean of the per-class Average Precision computed at a bounding-box overlap threshold of 50%. AP is the area under the precision-recall curve; mAP50 averages this across all classes:

$$\text{mAP50} = \frac{1}{N}\sum_{i=1}^{N} AP_{i@50}$$

mAP50-95 extends mAP50 by computing AP (Average Precision) at ten evenly spaced IoU thresholds from 0.50 to 0.95 (`torch.linspace(0.5, 0.95, 10)`) and averaging the result across thresholds and classes [4]. It penalises imprecise bounding box localisation more heavily than mAP50.

Recall is the primary metric for this project. A missed mast cell (false negative) causes the Atypisch fraction to be underestimated, potentially placing a patient below the WHO 25% diagnostic threshold and delaying diagnosis of SM. A false positive is less consequential for two reasons: the clinical workflow includes a manual review step where false alarms can be dismissed, and the diagnostic impact of a false positive depends on how far the true ratio is from the 25% threshold. In a patient with a strongly atypical cell population, a moderate false positive rate does not change the clinical conclusion because the Atypisch fraction remains clearly above the threshold. False positives become clinically relevant primarily in borderline cases where the ratio is close to 25%, and it is precisely in those cases that the manual review step is most likely to be exercised.

Two per-class recall metrics are tracked separately throughout the experiments: R_Atypisch (recall for the Atypisch class) and R_Normal (recall for the Normal class). Both are clinically necessary because the WHO criterion depends on the ratio of Atypisch to total detected mast cells. Undercounting Normal cells inflates this ratio, potentially triggering a false-positive SM classification in borderline patients. Tracking both classes independently, rather than reporting a single aggregate recall, is a direct consequence of aligning the evaluation to the diagnostic question rather than to standard object detection benchmarking practice. Published studies in adjacent domains, such as general white blood cell subtype classification, typically optimize for aggregate mAP or F1 across all classes [3]; this thesis treats R_Normal as a primary metric in its own right because the WHO threshold makes the Normal count as diagnostically load-bearing as the Atypisch count. R_Normal received particular experimental attention because Normal is the minority class in the training corpus, consistently exhibiting lower recall and higher cross-fold variability than R_Atypisch.

mAP50 is used as the standard benchmark for run-to-run comparison across the experimental timeline, providing a single-number summary that captures both detection and classification quality. mAP50-95 is reported where available but assigned less interpretive weight; localisation precision at high IoU thresholds is clinically less relevant than detection recall for this task. Precision is monitored throughout but treated as a secondary metric.

UI Creation

Gradio Prototype

An initial web-based interface was developed using Gradio to support early-stage result visualization and model evaluation during development. The prototype provided two functional tabs. The Analysis tab ran batch inference on image directories and displayed tiles with YOLO detection overlays (Figure 1), producing labeled output images. The Research tab supported ground-truth versus prediction comparison (Figure 2, Figure 3), applying a lower confidence threshold to maximize recall sensitivity during evaluation. A statistics dashboard (Figure 4) aggregated per-class detection counts and displayed the Atypisch fraction relative to the WHO 25% threshold. The Gradio prototype served its purpose as a rapid development tool and was later superseded by the desktop application.

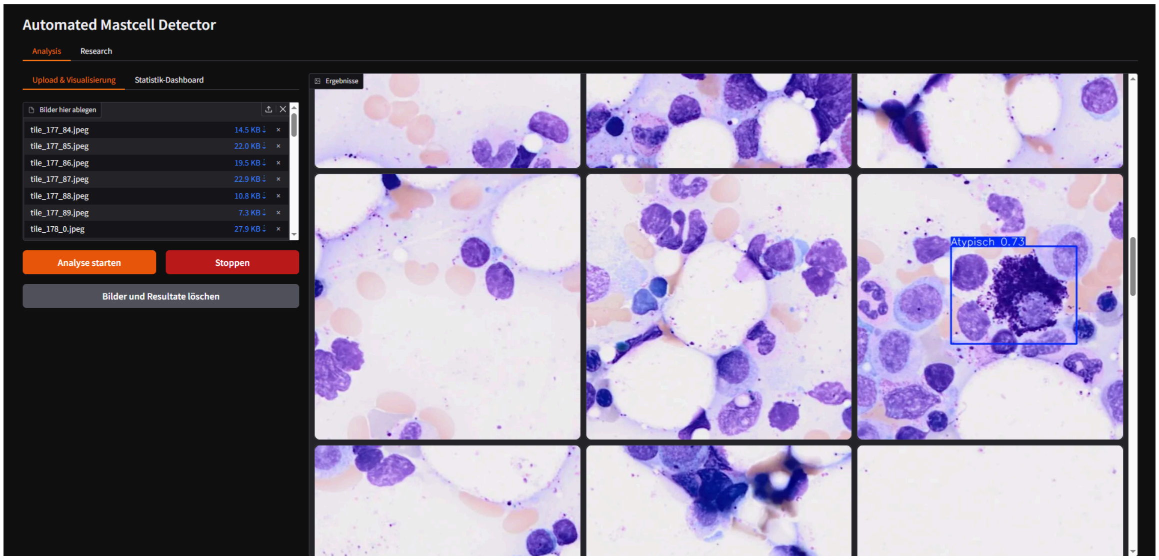

Figure 1: Analysis tab of the Gradio prototype showing uploaded image tiles with YOLO detection overlays.

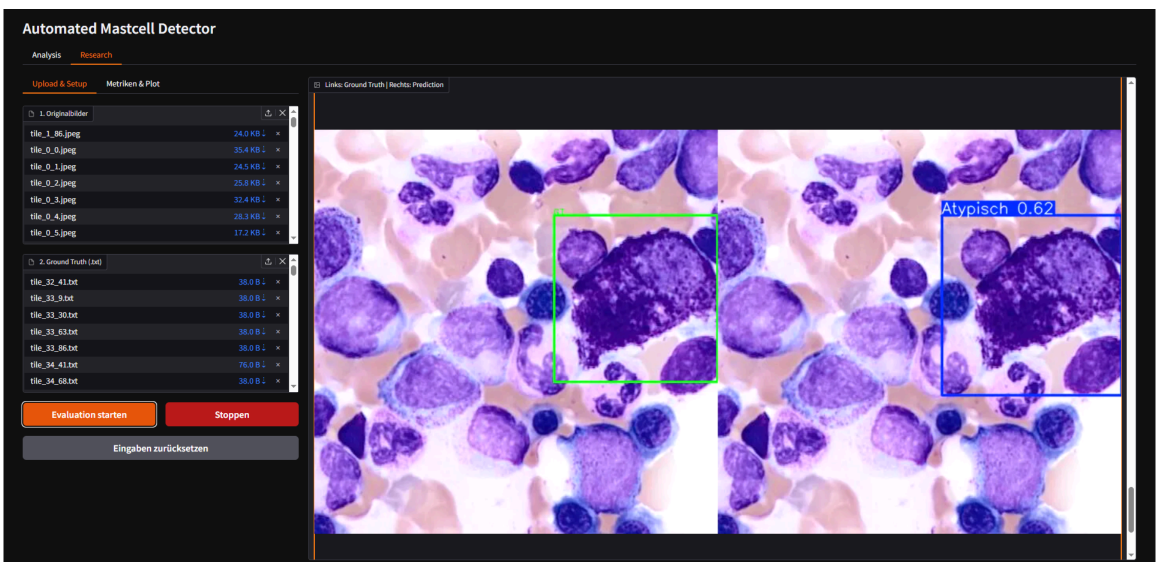

Figure 2: Research tab showing ground-truth annotations alongside model predictions for direct visual comparison.

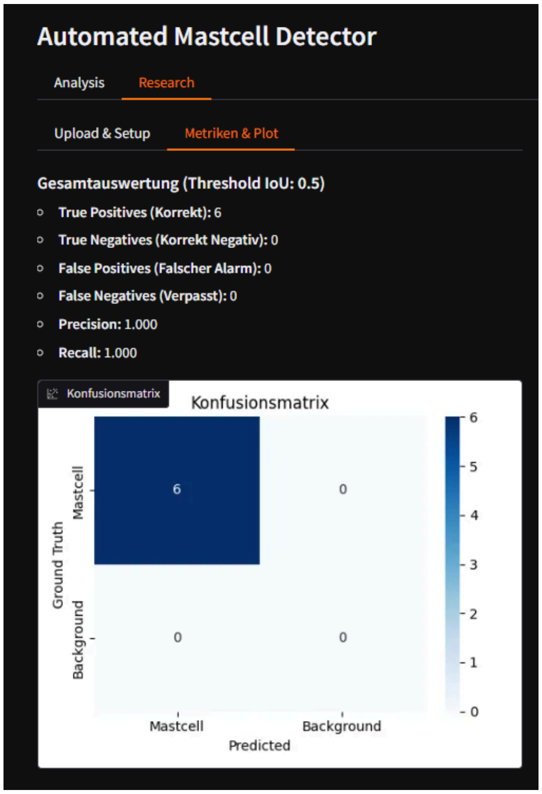

Figure 3: Research tab metrics output: precision-recall curves and performance plots generated from the ground-truth evaluation run.

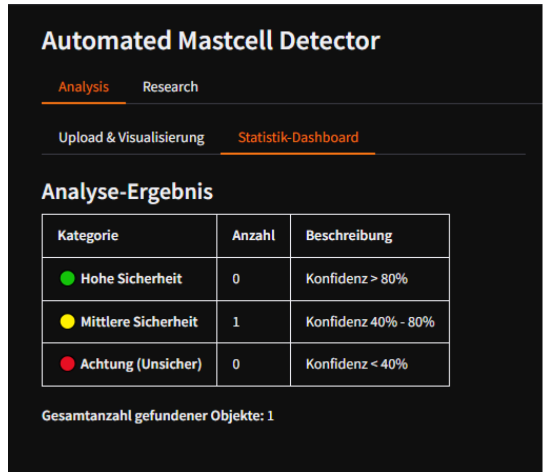

Figure 4: Statistics dashboard of the Gradio prototype showing per-class detection counts and the Atypisch fraction relative to the WHO 25% diagnostic threshold.

Desktop Application

As the requirements for clinical deployment became clearer, it became apparent that a browser-based interface was not adequate for the actual usage context at USZ. A full patient slide scan produces approximately 20,000 image tiles, and a Gradio-based interface was not designed for streaming inference at that scale. Cinderella, the cell classification software already in use at USZ, was considered as an integration target, but embedding the pipeline into an existing hospital codebase within the thesis timeline was not feasible: the time required to understand a production clinical codebase combined with the security validation and change management processes required for hospital software integration extended well beyond the project scope.

A native desktop application (MastCellDetector) was developed as the primary clinical deliverable. The application was built with PyQt and packaged using PyInstaller into a self-contained executable that requires no Python environment on the target machine. While the primary deployment target is Windows workstations at USZ, PyInstaller equally supports Linux builds, making the same application distributable on Linux servers or HPC systems.

The intended workflow for a USZ analyst proceeds as follows. The analyst opens a local folder containing a patient's image tiles. The sidebar provides controls for confidence threshold, IoU non-maximum suppression threshold, and hardware selection (GPU with automatic VRAM-based batch size recommendation, or CPU fallback). Running inference streams YOLO-format label files to disk alongside each image in real time, so results are preserved if the run is interrupted.

The gallery view (Figure 5) displays all images with filter tabs for All images, With detections (Figure 6), No detection, Low confidence, Atypisch only, and Normal only. This allows a haematologist to navigate directly to the cases most likely to need manual review rather than scrolling through tens of thousands of tiles.

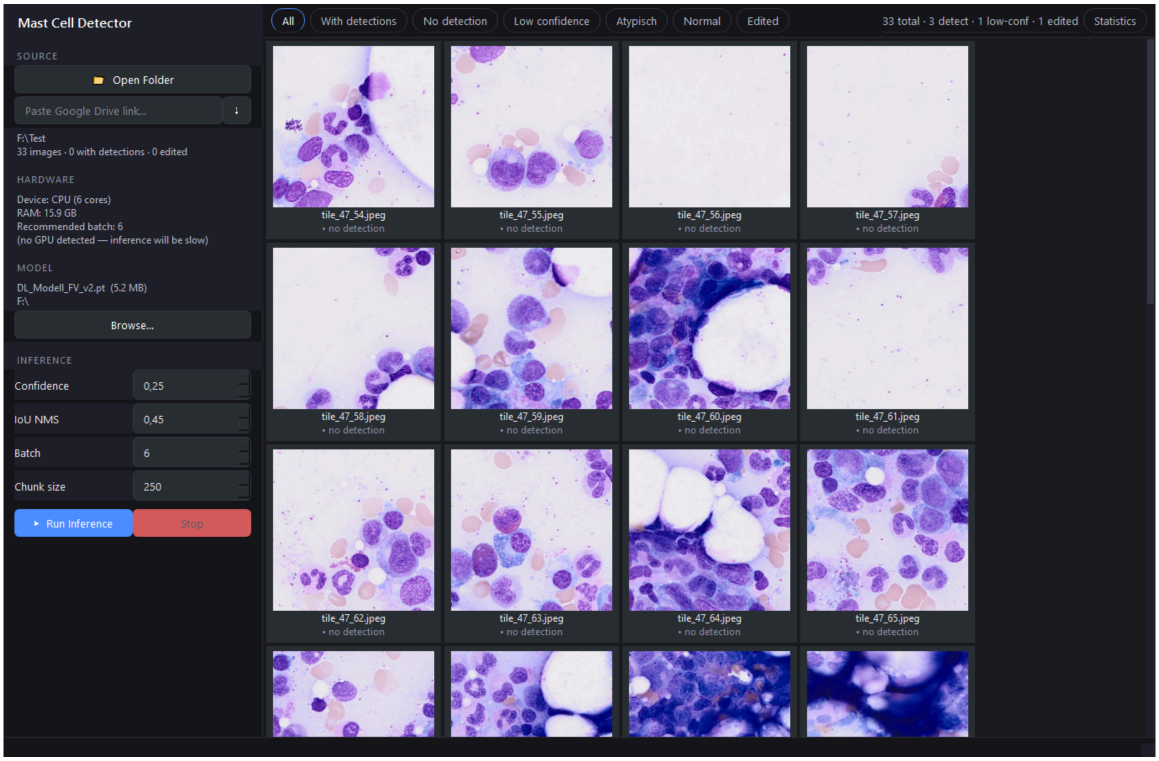

Figure 5: Gallery view of the desktop application showing all image tiles for a patient folder with sidebar controls for inference configuration.

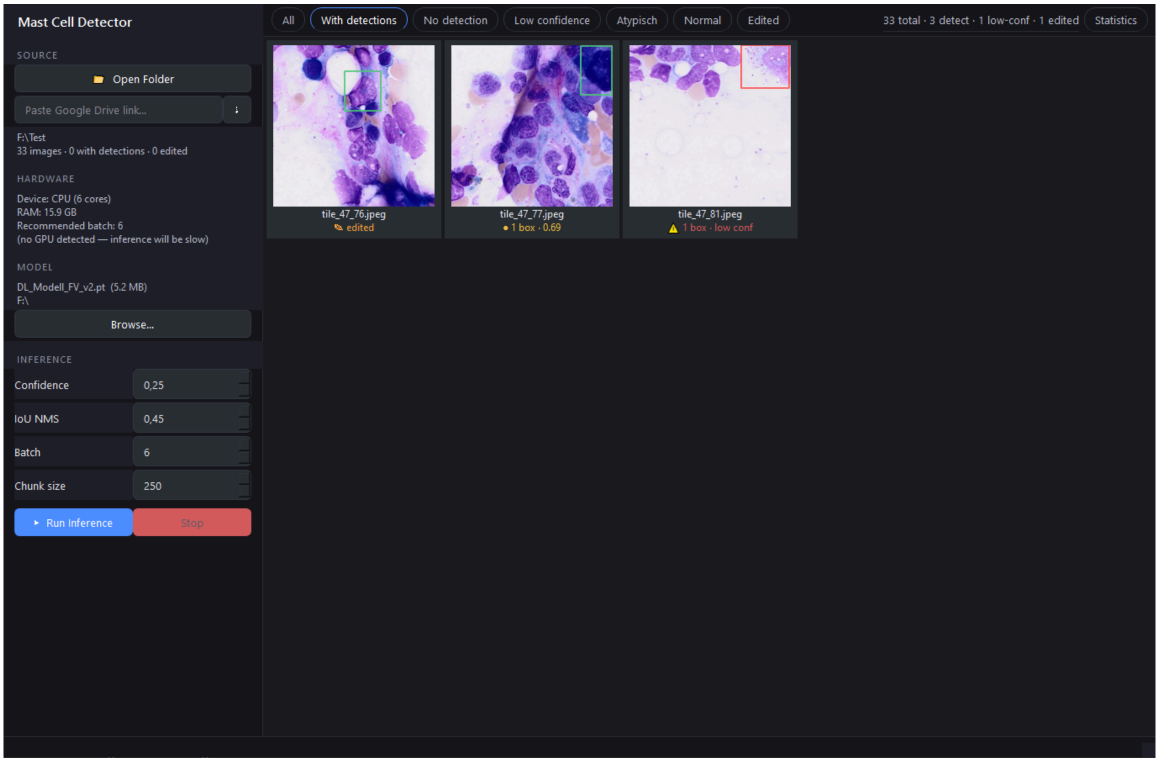

Figure 6: Gallery filtered to tiles with detections, allowing rapid navigation to images where the model found mast cells.

An annotation editor is embedded in the gallery. Clicking any thumbnail opens the image with interactive bounding boxes (Figure 7): boxes can be repositioned by dragging, resized by dragging corner handles, and deleted with the Delete key. New boxes are drawn by pressing A and clicking and dragging a region, with class selected from a dropdown. Changes are saved to the corresponding label file with Ctrl+S or automatically on navigation to the next image. When a box is deleted as a false positive, an empty label file is written, marking the tile as a confirmed negative for any future retraining cycle. Edited files are protected from being overwritten by subsequent inference runs.

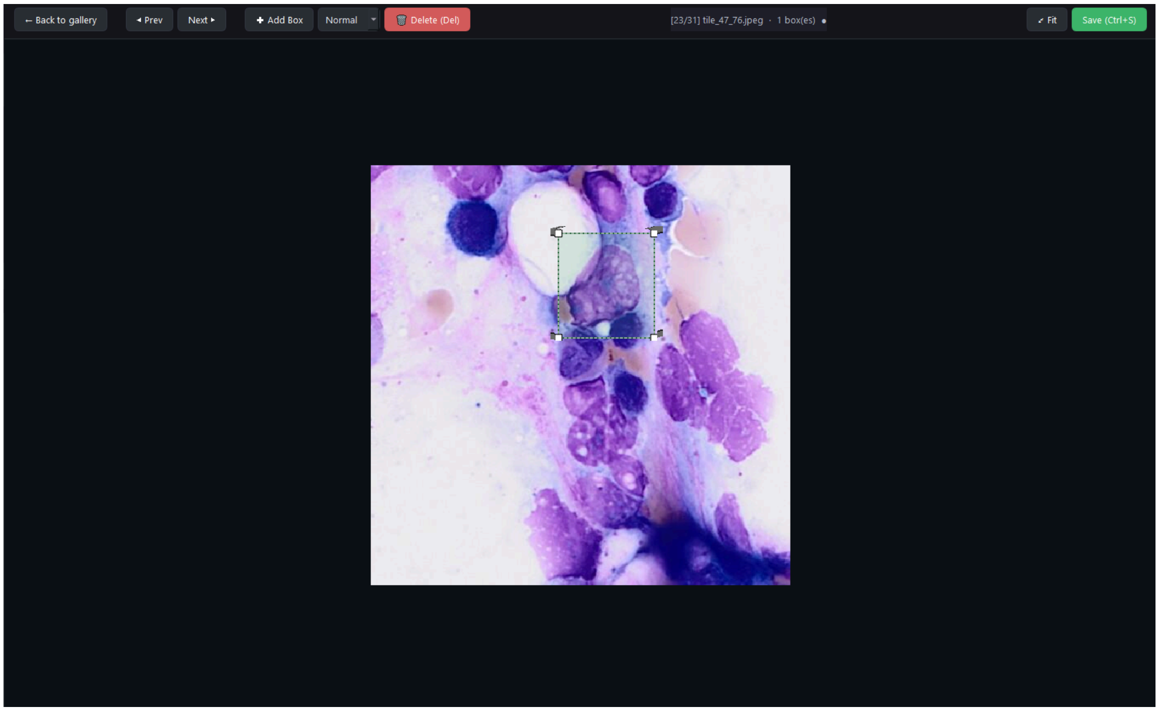

Figure 7: Annotation editor with interactive bounding boxes; boxes can be added, moved, resized, or deleted directly in the application without external tooling.

The Statistics tab (Figure 8) aggregates per-image detection counts into a patient-level summary that displays the total detected mast cells, the Atypisch fraction as a percentage, and a visual marker relative to the WHO 25% diagnostic threshold. This summary can be exported for clinical documentation or further analysis.

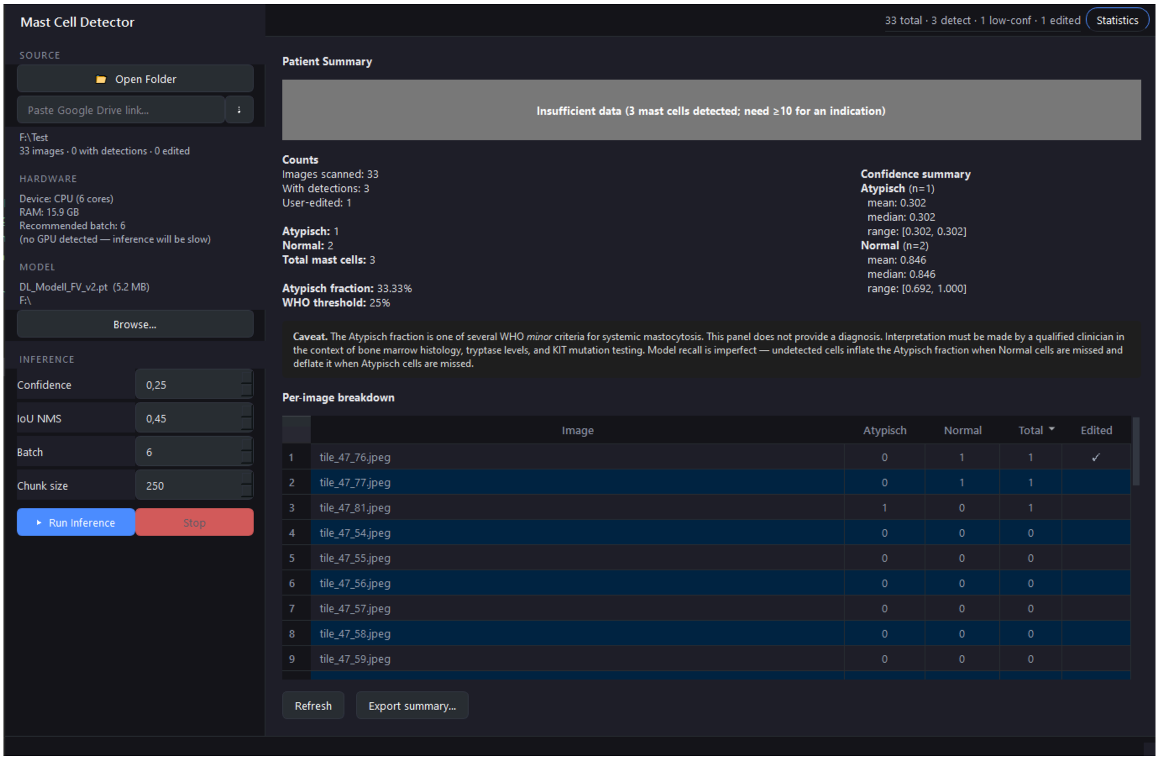

Figure 8: Statistics tab aggregating per-image detections into a patient-level summary with the Atypisch fraction displayed against the WHO 25% diagnostic threshold.

Implementation in USZ Infrastructure

The primary deliverable of this project is the MastCellDetector desktop application, distributed as a self-contained Windows directory. The application operates entirely offline after installation, which satisfies a practical requirement for deployment in hospital network environments where internet access from clinical workstations is restricted or prohibited.

The model shipped with the final version of the desktop application will be DL_Modell_FV_v3.pt, trained on the complete P1-P53 corpus using the production recipe established across Phases 3 and 4. This model was not yet available at the time of writing.

The annotation editing and label export functionality in the application also addresses a secondary deliverable: the confirmed false positives and corrected labels produced during clinical use feed directly back into the training pipeline as the next generation of hard-negative examples, continuing the iterative annotation strategy established from Phase 1 onward.

The Linux build capability opens a second deployment path beyond the clinical workstation. A Linux build of MastCellDetector can be deployed on a computational server or HPC node at USZ, allowing the tool to be integrated directly into the institution's scan-to-inference-to-evaluation pipeline: whole-slide images are scanned, tiles are passed through the model at server-side GPU throughput, and the resulting label files and statistics export are produced without requiring a Windows workstation in the loop. This makes the same application suitable for both interactive clinical review and automated batch processing, depending on the deployment context. The source code of the application is publicly available [5].

3. Results

The following training hyperparameters are referenced throughout this section:

- **`lr0`** — initial learning rate. Controls the step size at the start of fine-tuning; too high a value causes catastrophic forgetting of pretrained weights.
- **`freeze`** — number of backbone layers frozen during training. `freeze=10` locks the YOLO11n backbone and trains only the detection head; `freeze=0` allows full fine-tuning.
- **`degrees`** — maximum rotation angle for on-the-fly augmentation. Increases orientation diversity in the training distribution.
- **`dfl`** — Distribution Focal Loss weight. Scales the bounding-box regression loss relative to the classification loss; higher values shift the model toward localisation accuracy.
- **`cls`** — classification loss weight. Scales the class-prediction loss; used here to compensate for class imbalance between Atypisch and Normal.
- **`mosaic`** — probability of mosaic augmentation (four tiles composited into one image). Disabled (`mosaic=0.0`) throughout because mosaic scrambles single-cell 512 px crops.
- **`flipud`** / **`fliplr`** — vertical and horizontal flip augmentation probabilities.
- **`FP_NEG_OVERSAMPLE`** — repeat factor for confirmed false-positive tiles in the training split. Values above 1 up-weight hard negatives.
- **`BG_RATIO`** — ratio of unannotated background tiles added as negatives per training image.

3.1. Model Results and Statistical Evaluation

Performance metrics for Phase 3 and Phase 4 runs are reported from the test split of each LOPO fold. Phase 1 and Phase 2 results derive from held-out validation folds under 5-fold stratified cross-validation and are not directly comparable to Phase 3–4 LOPO estimates, for the reasons described in the Model Testing section. All values quoted in this section refer to cross-fold means unless stated otherwise.

Phase 1. The P1-baseline achieved validation mAP50-95 = 0.649 ± 0.131, precision = 0.768 ± 0.117, and recall = 0.719 ± 0.178. Cross-fold variance was high: Fold 1 reached mAP50-95 = 0.805 while Fold 5 collapsed at epoch 29 (recall = 0.432). No per-class breakdown was available as P1 contained only Atypisch annotations.

Phase 2. The 3×4 factorial DoE grid (11 cells, FP oversampling ∈ {1×, 2×, 3×} crossed with background ratio ∈ {0, 1, 2, 3}) found no statistically distinguishable difference across compositions: cell-level mAP50-95 variation was smaller than the within-cell cross-fold standard deviation. Background tiles decreased performance at every FP oversampling level. The fp=1, bgr=0 configuration (P2-H) was the grid winner at every metric, achieving mAP50-95 = 0.494 ± 0.166, precision = 0.934 ± 0.057, and recall = 0.537 ± 0.210.

Phase 3. The structural pivot to 8-fold LOPO on 8 patients with on-the-fly per-epoch augmentation produced the largest single performance step in the project. P3-A achieved mAP50-95 = 0.718 ± 0.137, precision = 0.764 ± 0.138, and recall = 0.840 ± 0.122, a 30.3 percentage point improvement in recall over P2-H (0.537). The cross-fold standard deviation dropped from 0.166 to 0.137 despite the harder evaluation protocol.

The tinkering matrix (P3-B, 14 of 24 cells completed) identified `degrees=5` rotation augmentation as the dominant training-time lever. The winning configuration (`degrees=5, dfl=1.5, freeze=10, label_smoothing=0.0`) achieved mAP50-95 = 0.741 ± 0.122, precision = 0.812 ±

0.153, recall = 0.866 ± 0.131, R_Atypisch = 0.892 ± 0.173, and R_Normal = 0.840 ± 0.176. `dfl=2.0` consistently regressed R_Normal relative to `dfl=1.5` across all degree and freeze combinations; `freeze=0` underperformed `freeze=10` in every recall-relevant slice; label smoothing was empirically inert (bit-identical fold metrics at `ls=0.0` and `ls=0.1`). These findings defined the production recipe (`degrees=5, dfl=1.5, freeze=10, lr0=0.001, cls=1.0`) adopted for all subsequent phases.

The automated hyperparameter tuner (P3-C, 30 iterations × 50 epochs, single P4 holdout, mAP50-95 fitness) converged on `degrees=0.0` and `dfl=2.47`. The production recipe from P3-B was retained.

The model-size comparison (P3-E, yolo11l, 43 M parameters) produced mAP50-95 = 0.745 ± 0.095, precision = 0.778 ± 0.134, recall = 0.785 ± 0.103, and R_Normal = 0.695 ± 0.144, the lowest R_Normal across all runs and 14.5 percentage points below P3-B.

Phase 4. Adding patients P9–P15 (424 new images, 293 new Normal annotations) expanded the corpus to 1,564 images and 15 LOPO folds. P4-A (degrees=0, the production recipe with the rotation variable inadvertently unset) achieved mAP50-95 = 0.793 ± 0.106, precision = 0.851 ± 0.097, recall = 0.904, R_Atypisch = 0.913, and R_Normal = 0.890. The R_Normal gain from corpus expansion (+13.6 pp over P3-A) exceeded the largest single hyperparameter gain observed in Phase 3 (+4.5 pp from `degrees=5` in P3-B).

P4-B (full production recipe with `degrees=5` explicitly set) produced mAP50-95 = 0.789 ± 0.118, precision = 0.826 ± 0.131, recall = 0.895 ± 0.132, R_Atypisch = 0.900 ± 0.174, and R_Normal = 0.882 ± 0.193. All P4-A vs P4-B deltas were within 1.6 percentage points. The freeze ablation (P4-C, `freeze=0`) produced mAP50-95 = 0.782 ± 0.124, precision = 0.863 ± 0.087, recall = 0.893 ± 0.135, R_Atypisch = 0.896 ± 0.172, and R_Normal = 0.882 ± 0.193, all within 0.4 percentage points of P4-B. P4-B is the primary reported result for the P1–P15 phase, as all training parameters are explicitly set.

Phase 5. Phase 5 evaluated the pipeline at full deployment scale across all 35 annotated patients from the 53-patient corpus. Two protocols were applied: a 35-fold LOPO run (P5-A) and a grouped 5-fold CV (P5-B). P5-B is the primary generalization result; P5-A is an optimistic upper bound documented for methodological completeness.

P5-A: 35-fold LOPO (optimistic upper bound). The production recipe was applied across all 35 annotated patients. Of the 35 folds, 34 completed within the 48-hour SLURM allocation; Fold 17 (P37 holdout) was retrained independently and evaluated in a post-hoc inference pass (see Section).

Across 35 folds, post-hoc evaluation reported mAP50-95 = 0.858 ± 0.120, precision = 0.925 ± 0.128, recall = 0.951 ± 0.084, R_Atypisch = 0.920 ± 0.129 (n=20 folds with Atypisch ground truth), and R_Normal = 0.941 ± 0.119 (n=29 folds with Normal ground truth). Of the 35 folds, 20 reported mAP50 at the metric ceiling (0.995), corresponding to patients from the P16–P53 acquisition batch who contributed only one to three annotated images each. On a one-image test set, a single correct detection produces recall = 1.000 regardless of model generalization capacity, inflating the aggregate mean.

P5-B: Grouped 5-fold CV (primary Phase 5 result). The 35 annotated patients were partitioned into five fixed groups, each serving as the test holdout exactly once. Every group contains at least one Atypisch-annotated and one Normal-annotated patient; the two largest patients (P1:

428 images, P2: 417 images) each anchor a separate group; test-image counts are balanced across groups (315–443 images per group). All five folds used the confirmed production recipe.

Across five groups, mAP50-95 = 0.748 ± 0.079, precision = 0.864 ± 0.022, recall = 0.853 ± 0.087, R_Atypisch = 0.885 ± 0.035, and R_Normal = 0.821 ± 0.165. Per-group metrics are in Section . G3 achieved R_Normal = 0.981, the highest across all multi-patient evaluation folds in the project.

Phase 6. A G1–G4 ablation run re-ran four-fold grouped CV on G1–G4 only, with G5 excluded from training, validation, and negative sets. Results against the corresponding P5-B G1–G4 folds: mAP50-95 = 0.738 (P5-B G1–G4: 0.740), Recall = 0.849 (+0.006), R_Atypisch = 0.897 (+0.009), R_Normal = 0.801 (+0.004). All deltas fall within cross-fold noise, confirming that training without G5 is feasible with no substantial performance loss.

This finding raised the question of whether G5 could be withheld entirely as a calibration cohort. That approach was ruled out: to make use of all G5 patients for calibration, inference would need to be run across the full slide set of each G5 patient (roughly 20,000 tiles per patient, approximately 380′000 tiles in total), which is impractical for a single calibration pass. The alternative, withholding all of G5 from training but calibrating on only one of its patients, would discard image information from the remaining G5 patients without benefit. Instead, a single patient was withheld from training: P11, selected because its 17 Atypisch and 20 Normal annotations provide near-equal class balance, the most informative single-patient confidence reference in the corpus. All remaining patients, including the rest of G5, were retained in training. The final deployment model was trained on P1–P53 minus P11 (1,832 images, 1,459 Atypisch, 491 Normal, 3.0:1 ratio) using the production recipe unchanged.

Table 2 summarizes the key milestone runs across all phases. The complete experiment record, including corrupted runs and all 11 DoE grid cells, is maintained in `hpc/experiment_log.md` [6]. R_Normal for P3-A is reported as the mean over the four Normal-rich holdout folds only (P5, P6, P7, P8, each with at least 6 Normal instances); folds with fewer than 3 Normal instances in the test set (P1: 2, P3: 4) are excluded from that average because a single missed cell produces metric swings of 50 and 25 percentage points respectively, making those values dominated by small-sample noise rather than model behaviour.

Table 2: Milestone run progression summary across all experimental phases. All metrics shown as mean ± std across folds; base model is yolo11n throughout except P3-E (yolo11l, 43 M parameters). CV-codes: S = stratified k-fold, L = leave-one-patient-out (LOPO), CV = grouped cross-validation; prefix = fold/group count. *P3-A per-class recall values are restricted-subset means (see text); fold-level std not available.

Run	CV	mAP50-95	Prec.	Recall	R_Atyp	R_Norm
P1-base	5f-S	0.649 ± 0.131	0.768 ± 0.117	0.719 ± 0.178	—	—
P2-C	5f-S	0.344 ± 0.031	0.880 ± 0.017	0.397 ± 0.022	—	—
P2-H	5f-S	0.494 ± 0.166	0.934 ± 0.057	0.537 ± 0.210	—	—
P3-A	8f-L	0.718 ± 0.137	0.764 ± 0.138	0.840 ± 0.122	0.884*	0.861*
P3-B	**8f-L**	**0.741 ± 0.122**	**0.812 ± 0.153**	**0.866 ± 0.131**	**0.892 ± 0.173**	**0.840 ± 0.176**
P3-E	8f-L	0.745 ± 0.095	0.778 ± 0.134	0.785 ± 0.103	0.875 ± 0.169	0.695 ± 0.144
P4-A	15f-L	0.793 ± 0.106	0.851 ± 0.097	0.904 ± 0.120	0.913 ± 0.110	0.890 ± 0.194
P4-B	**15f-L**	**0.789 ± 0.118**	**0.826 ± 0.131**	**0.895 ± 0.132**	**0.900 ± 0.174**	**0.882 ± 0.193**
P4-C	15f-L	0.782 ± 0.124	0.863 ± 0.087	0.893 ± 0.135	0.896 ± 0.172	0.882 ± 0.193
P5-A	35f-L	0.858 ± 0.120	0.925 ± 0.128	0.951 ± 0.084	0.920 ± 0.129	0.941 ± 0.119
P5-B	**5g-CV**	**0.748 ± 0.079**	**0.864 ± 0.022**	**0.853 ± 0.087**	**0.885 ± 0.035**	**0.821 ± 0.165**
P6	—	pending	—	—	—	—

P1-base established the initial proof of concept on a single patient. P2-C identified the learning rate mismatch that suppressed recall in all DL_Modell_FV runs before the fix; P2-H is the Phase 2 reference after the learning rate and freeze correction. P3-A marks the single largest structural improvement in the project. P3-B confirmed the production recipe through the 8-fold tinkering matrix and is the first fully validated configuration (bold). P3-E is a deliberate negative result bounding model-size choices. P4-A documents the corpus expansion gain from P9–P15; P4-B is the primary result for the P1–P15 phase with all recipe parameters explicitly set. P4-C is the freeze ablation. P5-A is the optimistic upper bound at full corpus scale; P5-B is the primary cross-validated generalization result. P6 had not yet completed evaluation at the time of writing.

3.2. Quantitative and Qualitative Evaluation

Per-Fold Breakdown (P5-A)

Table 3 presents the complete 35-fold post-hoc evaluation results for P5-A. Each row corresponds to one annotated patient held out from training entirely; all metrics are evaluated on that patient's images exclusively. Post-hoc evaluation was applied uniformly across all 35 folds because Fold 17 (P37 holdout) was killed by the 48-hour SLURM wall-time limit and retrained independently; running the unified post-hoc pass over all folds ensures a consistent result set without mixed evaluation protocols (see Section). Patients are listed in numeric order.

Table 3: P5-A per-fold test results (35-fold LOPO post-hoc evaluation, full P1–P53 annotated corpus). GT Atyp / GT Norm = ground-truth cell instance counts in the held-out test split. "—" indicates the class was absent from that test split and is excluded from the per-class mean. R_Atyp mean over n=20 folds; R_Norm mean over n=29 folds.

Patient	GT Atyp	GT Norm	mAP50-95	mAP50	Prec.	Recall	R_Atyp	R_Norm
P1	482	2	0.610	0.813	0.577	0.973	0.946	1.000
P2	449	6	0.600	0.716	0.721	0.712	0.924	0.500
P3	60	4	0.736	0.906	0.785	0.931	0.911	0.951
P4	122	6	0.675	0.806	0.754	0.796	0.926	0.667
P5	3	18	0.652	0.760	0.595	0.740	0.667	0.814
P6	0	6	0.846	0.995	1.000	1.000	—	1.000
P7	8	12	0.809	0.975	0.929	0.917	1.000	0.833
P8	0	24	0.816	0.986	0.914	0.958	—	0.958
P9	4	195	0.600	0.622	0.635	0.722	0.500	0.944
P10	5	20	0.909	0.991	0.987	0.906	1.000	0.812
P11	17	20	0.863	0.964	0.968	0.887	0.941	0.832
P12	133	0	0.832	0.984	0.962	0.952	0.952	—
P13	0	7	0.920	0.995	1.000	1.000	—	1.000
P14	5	3	0.933	0.995	0.978	1.000	1.000	1.000
P15	0	68	0.949	0.991	0.984	0.985	—	0.985
P16	22	1	0.834	0.994	1.000	0.977	0.955	1.000
P17	69	0	0.832	0.992	0.945	0.994	0.994	—
P18	81	1	0.948	0.993	0.934	0.946	0.893	1.000
P24	1	0	0.796	0.995	1.000	1.000	1.000	—
P28	1	0	0.995	0.995	1.000	1.000	1.000	—
P29	1	0	0.896	0.995	1.000	1.000	1.000	—
P31	0	1	0.995	0.995	1.000	1.000	—	1.000
P37	0	2	0.896	0.995	1.000	1.000	—	1.000
P39	0	1	0.896	0.995	1.000	1.000	—	1.000
P40	0	1	0.995	0.995	1.000	1.000	—	1.000
P43	0	3	0.940	0.995	1.000	1.000	—	1.000
P44	0	2	0.921	0.995	1.000	1.000	—	1.000
P45	0	1	0.995	0.995	1.000	1.000	—	1.000
P46	0	4	0.967	0.995	1.000	1.000	—	1.000

Patient	GT Atyp	GT Norm	mAP50-95	mAP50	Prec.	Recall	R_Atyp	R_Norm
P47	0	3	0.918	0.995	1.000	1.000	—	1.000
P48	7	0	0.857	0.995	1.000	1.000	1.000	—
P50	0	1	0.995	0.995	1.000	1.000	—	1.000
P51	5	90	0.700	0.765	0.740	0.900	0.800	1.000
P52	1	4	0.945	0.995	0.971	1.000	1.000	1.000
P53	0	5	0.952	0.995	1.000	1.000	—	1.000
Mean			**0.858**	**0.948**	**0.925**	**0.951**	**0.920** (n=20)	**0.941** (n=29)

P1 R_Normal = 1.000 based on 2 Normal ground-truth instances and is not a stable estimate. P12 has 133 Atypisch and 0 Normal ground-truth instances; R_Normal is absent from its test set.

Nineteen folds reported mAP50 at the ceiling value of 0.995, all corresponding to patients from the P16–P53 acquisition batch who contributed one to three annotated images each. On a one or two-image test set a single correct detection drives recall and mAP50 to their respective maxima regardless of model generalization capacity. These folds confirm that the model produces no catastrophic failures on any individual patient from the expanded corpus, but they carry no cross-patient generalization signal. The five patients with mAP50 below 0.820 (P2, P4, P5, P9, P51) represent the meaningful lower tail of the performance distribution at full corpus scale; their per-fold patterns are examined in Section .

Per-Group Breakdown (P5-B)

Table 4 presents the complete per-group test results for P5-B. Each row corresponds to one held-out patient group; all metrics are evaluated exclusively on images from patients in that group. The holdout column identifies the constituent patients; G5 contains the six large-corpus patients from P4 and P5 plus all 13 remaining small-corpus patients from P16–P53.

Table 4: P5-B grouped 5-fold CV split composition and test results (35 annotated patients, P1–P53 corpus). Train = annotated (positive) images; each fold additionally receives 3,773-4,595 negative tiles. Atyp/Norm = ground-truth label counts in the test split. All five groups have both classes present.

Group	Holdout patients	Train	Val	Test	Atyp	Norm	mAP50-95	mAP50	Prec.	Recall	R_Atyp	R_Norm
G1	P1, P13, P31	1219	214	436	482	10	0.649	0.766	0.850	0.746	0.904	0.588
G2	P2, P8, P37	1213	213	443	449	32	0.679	0.858	0.839	0.773	0.836	0.710
G3	P9, P12, P53	1315	231	323	137	200	0.813	0.913	0.882	0.932	0.884	0.981
G4	P3, P6, P15–P18, P48	1322	232	315	239	80	0.821	0.918	0.892	0.919	0.928	0.909
G5	P4, P5, P7, P10–P11, P14, P24, P28–P29, P39–P40, P43–P47, P50–P52	1289	228	352	169	189	0.777	0.905	0.854	0.896	0.873	0.918
Mean ± std							**0.748 ± 0.079**	**0.872 ± 0.064**	**0.856 ± 0.024**	**0.853 ± 0.087**	**0.885 ± 0.035**	**0.821 ± 0.165**

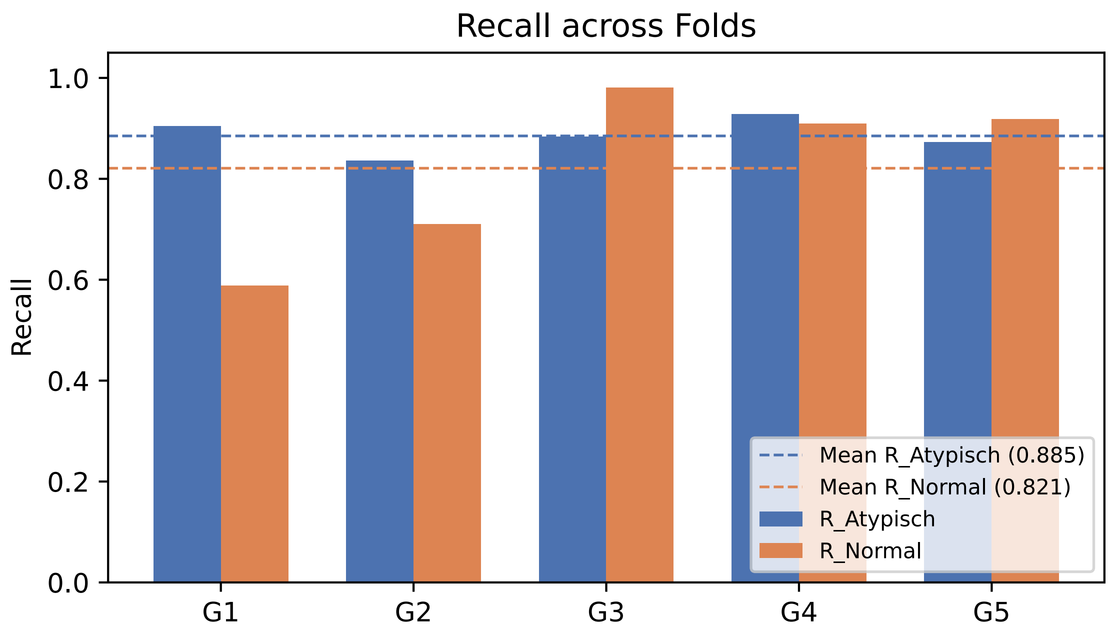

Figure 9: P5-B per-group recall by class. Bars show R_Atypisch (blue) and R_Normal (orange) for each held-out patient group. Dashed horizontal lines mark the cross-fold means (R_Atypisch = 0.885, R_Normal = 0.821).

3.3. Evaluation Visualization

Training convergence for P5-B is shown in Figure 10 for a representative fold (Fold 3, G3 holdout: P9, P12, P53; mAP50 = 0.913, recall = 0.932). The training and validation loss curves converge within approximately 150 epochs on the majority of folds, with YOLO's early-stopping criterion (patience = 50) triggering at the validation mAP50 plateau.

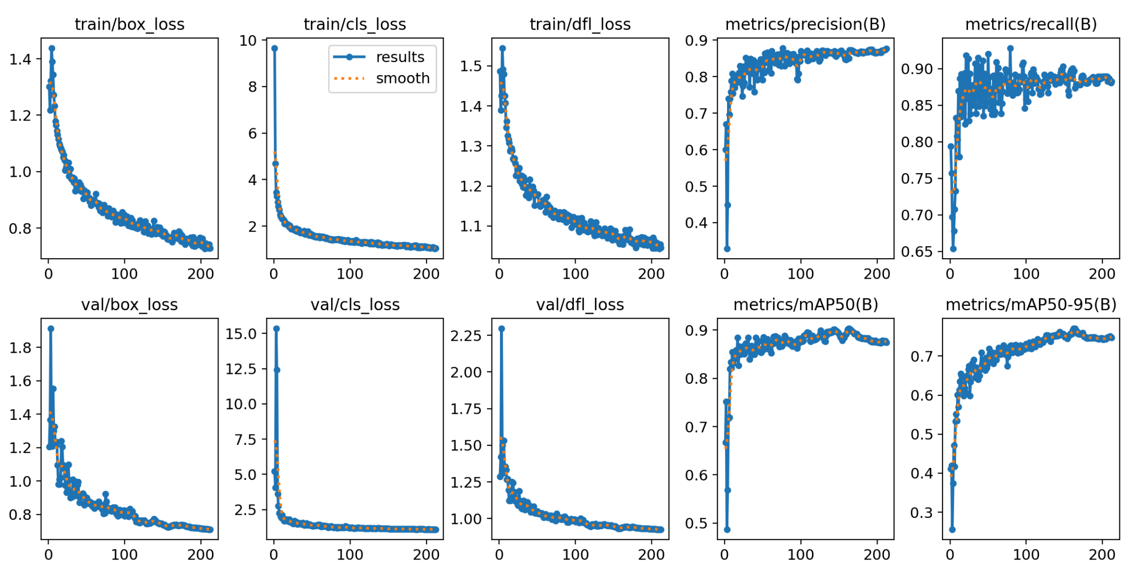

Figure 10: Training convergence for P5-B Fold 3 (G3 holdout: P9, P12, P53). Panels from top left: box loss, classification loss, and distribution focal loss for training and validation splits; precision, recall, mAP50, and mAP50-95 across epochs.

The normalized confusion matrix for the same fold is shown in Figure 11. Row normalization is applied because the Atypisch and Normal test-set sizes differ, making raw counts incomparable across classes; row-normalized values represent the fraction of each ground-truth class assigned to each predicted class. The main diagonal records 0.96 for Atypisch and 0.88 for Normal, indicating the fraction of each class correctly detected. Cross-class misclassifications between Atypisch and Normal are negligible at 0.01. The dominant off-diagonal entry is 0.91 in the background-to-Atypisch cell, reflecting the proportion of background tiles assigned to the Atypisch class at the evaluation threshold of 0.58.

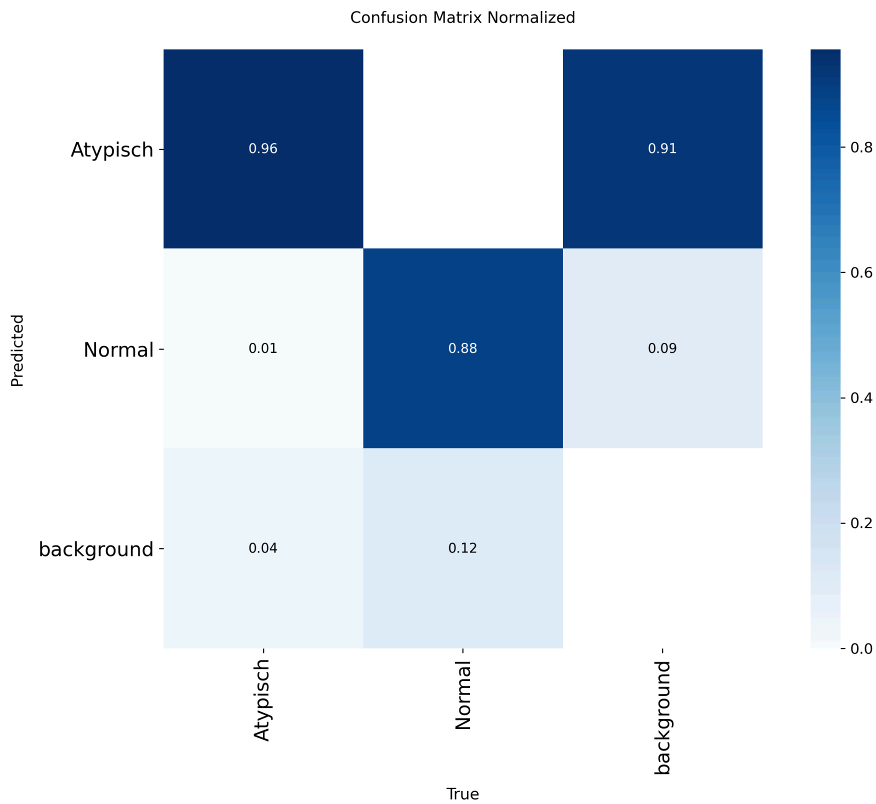

Figure 11: Normalized confusion matrix for P5-B Fold 3 (G3 holdout: P9, P12, P53). Rows are ground-truth classes; columns are predicted classes. Values are row-normalized fractions. The main diagonal entries are 0.96 (Atypisch) and 0.88 (Normal); cross-class error is 0.01; background-to-Atypisch is 0.91.

The precision-recall curve for the same fold is shown in Figure 12. Both classes trace high-precision curves across a broad recall range before the precision cliff. At the deployed threshold of 0.58, both curves are in their high-precision operating region.

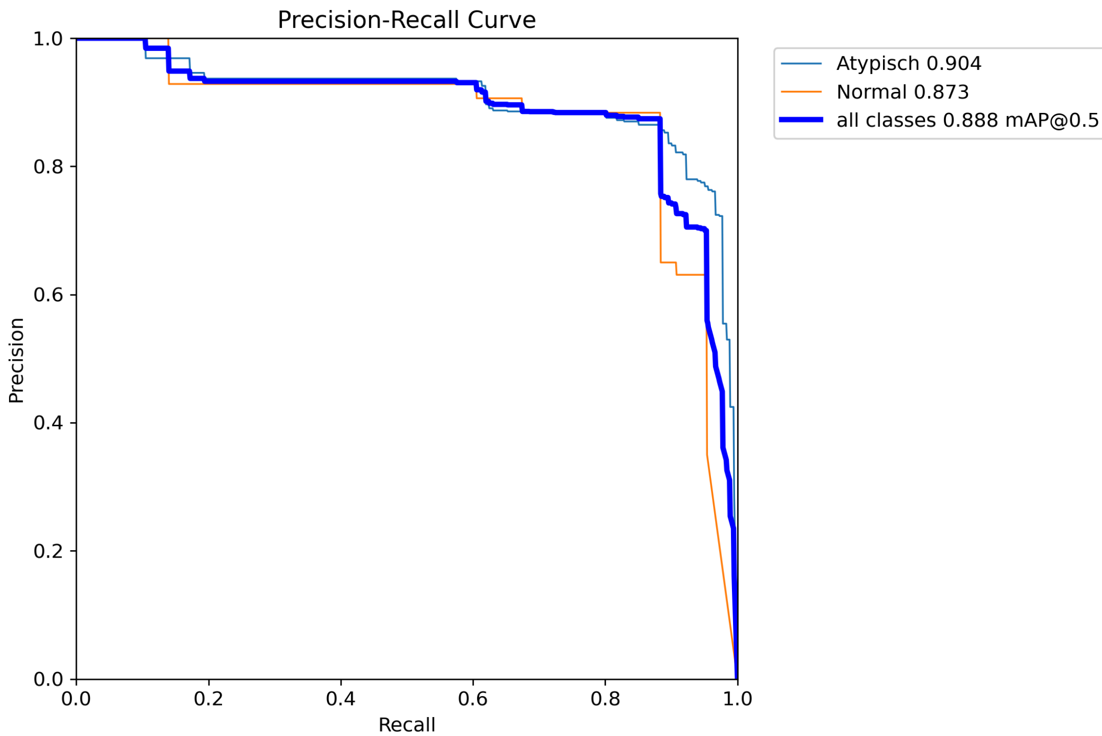

Figure 12: Box precision-recall curves for P5-B Fold 3 (G3 holdout) at IoU threshold 0.50. Atypisch (blue) and Normal (orange) curves illustrate the per-class discrimination capacity across confidence thresholds.

Figure 9 shows per-group R_Atypisch and R_Normal for all five P5-B folds alongside the cross-fold means.

Figure 13 compares all six evaluation metrics between P5-A and P5-B. Error bars show one standard deviation across folds for both evaluations; R_Atyp and R_Norm for P5-A are computed over the subset of folds where the respective class was present in the test set.

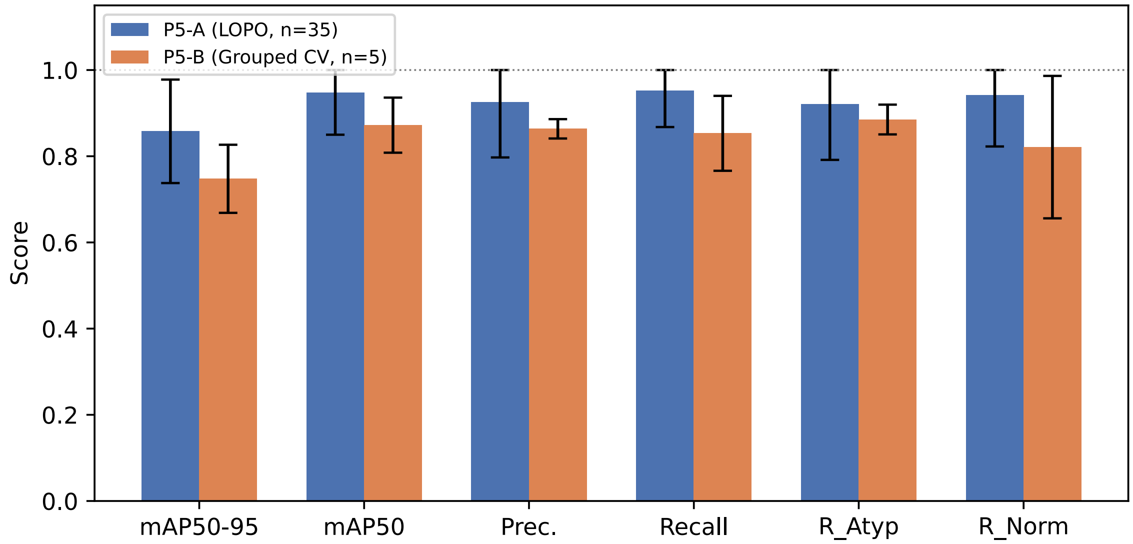

Figure 13: Metric comparison between P5-A (35-fold LOPO) and P5-B (grouped 5-fold CV). Error bars represent ±1 standard deviation across folds. R_Atyp and R_Norm for P5-A are computed over the 20 and 29 folds respectively where the class was present in the test set.

3.4. Real World Usage

Confidence Threshold Calibration

After Phase 1 training, an inference run was conducted on the complete first-patient slide set, approximately 20,000 tiles, with all training images excluded by filename. The resulting confidence distribution, shown in Figure 14, illustrates how the Phase 1 model distributed detection scores across the full tile set.

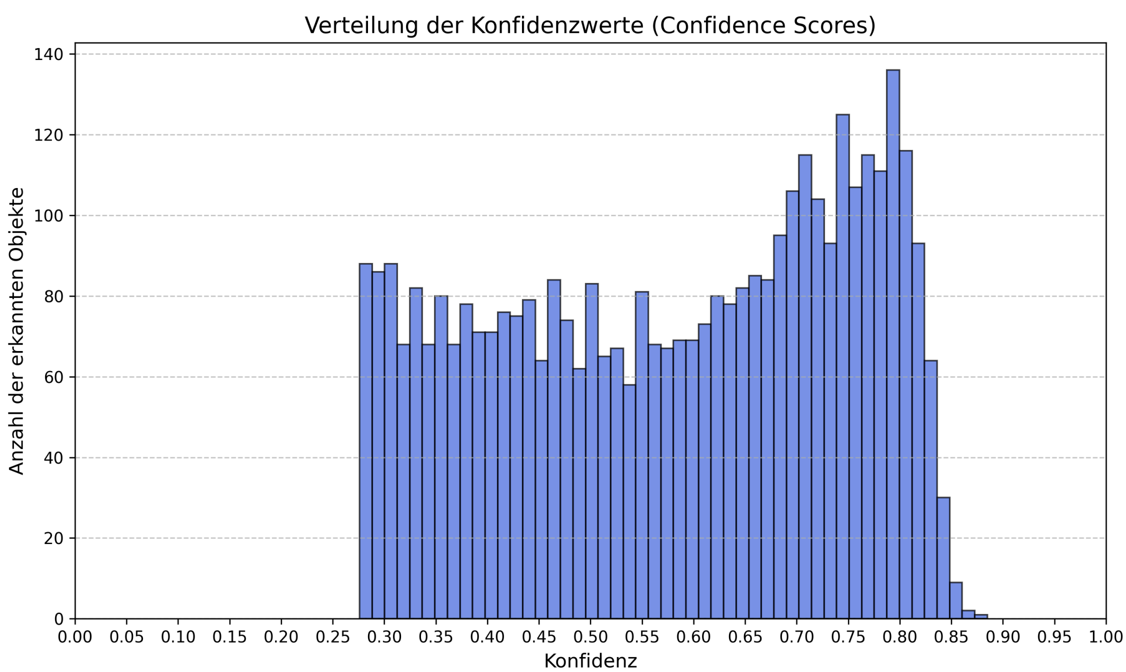

Figure 14: Confidence distribution of the Phase 1 model across the full P1 slide set (~20,000 tiles).

The distribution in Figure 14 shows a high-confidence detection peak and a broad low-confidence region. The deployment threshold of conf = 0.58 was calibrated from this distribution, sitting above the low-confidence bulk while retaining the high-confidence peak. The thresholds conf = 0.25 (Research tab) and conf = 0.276 (inference.py) are used to capture the full confidence distribution for downstream analysis.

Desktop Application Delivery

The MastCellDetector desktop application was delivered to USZ alongside the P3-D model weights as the primary clinical tool for patient-scale inference. The application runs inference on complete patient tile folders, streams YOLO label files to disk in real time, and presents results through the gallery and statistics interface described in the Methodology section. The Statistics tab produces the patient-level Atypisch fraction summary relative to the WHO 25% diagnostic threshold.

Phase 6 Full-Corpus Inference

Phase 6 applied the P5-B production model to the complete 53-patient slide archive via `hpc/inference.py` at a confidence threshold of 0.276, capturing the full detection score distribution for each patient. The inference run excluded all annotated training tiles by filename, operating exclusively on unannotated tiles. Per-patient detection counts and Atypisch fractions were recorded for all 53 patients, providing the first full-corpus inference output from a model trained on the complete annotated dataset.

4. Discussion

4.1. Interpretation of Results

The metric range across groups in P5-B is wide but structured. G1′s mAP50 of 0.766 is depressed by the P1 Normal fragility described throughout Phase 3 and Phase 4; it is not a Phase 5 regression. G2′s recall of 0.773 reflects that P2 and P8 are the two most Atypisch-dominant patients in the corpus (Atypisch:Normal ratio exceeding 60:1 for P2), making the G2 test set particularly sensitive to any missed Atypisch cell in terms of overall recall. G3, G4, and G5 each achieve recall above 0.89, consistent with the P4-B LOPO mean of 0.895 on the earlier 15-patient corpus. The convergence of G3–G5 with the P4-B baseline indicates that training on the expanded 30-patient set (35 minus holdout group) did not introduce distributional drift relative to the P1–P15 results; instead, it maintained performance while adding 20 new patients to the evaluation scope.

The grouped CV design is not equivalent to LOPO but preserves the essential patient-level holdout property: no image from any holdout patient appears in training within a given fold. The test-set sizes of 315–443 images per group are large enough to absorb individual missed detections without catastrophic recall swings, which is the core property that differentiates P5-B from P5-A as a generalization estimate. The wall-time cost of five folds (approximately 36 hours at MAX_PARALLEL=2) fits within the 96-hour HPC allocation that made 35-fold LOPO infeasible, making grouped patient CV the tractable approximation of LOPO at this corpus scale.

Per-Class Recall Analysis

The per-class recall gap between R_Atypisch and R_Normal narrowed from approximately 9 percentage points at the Phase 3 baseline (P3-A: R_Atypisch = 0.884, R_Normal = 0.795, computed from the P3-B tinkering matrix default cell) to 1.8 percentage points at the Phase 4 best run (P4-B: R_Atypisch = 0.900, R_Normal = 0.882). The narrowing reflects two simultaneous effects: a 12.9 percentage point absolute improvement in R_Normal driven primarily by patient P9′s 195 Normal annotations, and a 1.6 percentage point improvement in R_Atypisch from the additional Atypisch-dominant patients P10–P15. Neither effect alone explains the gap reduction; the combination of more Normal training signal and better cross-patient Atypisch diversity together narrowed the difference.

R_Normal is the more clinically constrained metric throughout the project. The R_Normal sequence across key milestones traces the cumulative contribution of each methodological intervention: 0.695 (P3-E, yolo11l), 0.795 (P3-A, on-the-fly augmentation and LOPO), 0.840 (P3-B, rotation augmentation), 0.890 (P4-A, patient P9 added), 0.882 (P4-B), 0.941 (P5-A, 35-fold LOPO), and 0.821 (P5-B grouped CV). The single largest R_Normal gain at any step (+9.5 pp from P3-B to P4-A) came from corpus expansion, not from hyperparameter adjustment, consistent with the Phase 2 DoE's controlled negative result on data composition within a fixed corpus. The P5-A figure of 0.941 is ceiling-inflated by the 19 folds containing one or two Normal test instances where a single correct detection guarantees R_Normal = 1.000; it should not be interpreted as a generalization estimate. The P5-B figure of 0.821 is the conservative generalization estimate: it evaluates across 35 patients including G1, whose test set contains only 10 Normal ground-truth instances, making it structurally more challenging than the P1–P15 cohort. G3 achieves R_Normal = 0.981, the highest value in any multi-patient fold across all phases. The trajectory confirms that improving rare-class recall in this domain requires more biologically diverse patients rather than better arrangements of existing data, and that at 35-patient scale the pipeline has acquired sufficient Normal-class coverage to achieve near-perfect recall in Normal-dominant test populations.

Fold-Level Structural Patterns

Several fold-level patterns in Table 3 reflect properties of the corpus rather than model behaviour.

P9 is the weakest informative fold. P9 contributes 4 Atypisch and 195 Normal instances to the test set. A single missed Atypisch cell moves R_Atypisch by 25 percentage points; P5-A recorded R_Atypisch = 0.500 (2 of 4 detected). The resulting mAP50 of 0.622 is the lowest across all 35 folds. P9′s extreme class asymmetry (4 Atypisch against 195 Normal) makes it the hardest fold by construction: the model trains on a broadly Atypisch-dominant corpus but must classify correctly in a patient where Normal cells outnumber Atypisch by nearly 50:1. The high R_Normal of 0.944 on the same fold confirms that the model handles the Normal-dominant slide correctly when P9 is withheld; the weakness is specific to the four rare Atypisch instances.

P1 recovered from its Phase 3–4 structural weakness. The P1 holdout fold, which produced mAP50 = 0.584 under the 14-patient P4-B training set, improved to mAP50 = 0.813 under the 34-patient P5-A training set. The annotation-regime gap between P1′s original `Pos_neg` directory structure and the CVAT-assisted P2–P53 annotations remains in place, but the expanded corpus provides enough additional Atypisch-dominant training material to partially compensate for the distributional mismatch. R_Atypisch = 0.946 and R_Normal = 1.000 (on 2 Normal instances) are both higher than in any prior phase under this holdout.

P2 and P5 show small-sample minority-class fragility. The P2 holdout (449 Atypisch, 6 Normal) produced R_Normal = 0.500 (3 of 6 detected); a single additional missed Normal cell would reduce this to 0.333. P5 (3 Atypisch, 18 Normal) recorded R_Atypisch = 0.667 (2 of 3 detected); one missed Atypisch shifts the value by 33 percentage points. Both figures are dominated by small-sample noise rather than systematic model error. P5′s R_Normal = 0.814 and P2′s R_Atypisch = 0.924 confirm that each patient’s majority class is detected reliably; the fragility is confined to the minority class within each specific test set.

Ceiling folds are structurally uninformative. The 19 folds at mAP50 = 0.995 correspond to patients from the P16–P53 batch who contributed one to three annotated images each. Perfect or near-perfect recall on a one-image test set is mathematically guaranteed when the single annotated cell is detected; these values confirm the absence of catastrophic failures but carry no information about cross-patient generalization capacity. Per-fold recall values throughout Table 3 are informative as lower bounds (for structurally fragile small-sample folds) and as ceiling observations (for the P16–P53 batch) rather than as stable point estimates.

Measurement Notes

A per-class recall extraction issue affected all LOPO runs through Phase 4. The utility function indexed Ultralytics’ per-class result arrays by position rather than by the `ap_class_index` field. When a holdout patient contained only one class and the model produced predictions for that class only, Ultralytics returned a one-element rather than a two-element result array, causing positional indexing to assign the single element to R_Atypisch and report R_Normal as NaN. This affected Fold 5 (P13 holdout) in all Phase 3 and Phase 4 runs; the true R_Normal for that fold is 1.000, confirmed from the per-fold confusion matrix. Affected folds are excluded from per-class cross-fold means and clearly marked in all fold tables.

For the P5-A post-hoc evaluation, the bug was fixed by examining per-patient ground-truth label files before calling `val()`: the label-derived GT class counts provided a reliable class-index lookup independent of Ultralytics' internal `ap_class_index` field. This fix allowed the 35-fold P5-A results to be extracted correctly without per-fold manual inspection; folds where only one class had ground truth instances were routed to the appropriate class slot by GT count rather than by position. The aggregate metrics mAP50 and overall Recall are computed internally by Ultralytics and are unaffected by this issue in all phases.

4.2. Clinical Relevance

4.3. Error Analysis

5. Conclusion

5.1. Summary of Key Findings

5.2. Limitations and Challenges

5.3. Future Research Directions

6. Declaration on AI Assistance

The author declares that the empirical research, data analysis, experimental design, and quantitative results presented in this thesis were generated independently and strictly remain the intellectual property of the author.

Artificial intelligence (AI) assistance was utilized solely for the purpose of language refinement, stylistic editing, and optimization of academic reading flow. Specifically, the AI models (Gemini 2.5 Pro, Claude Sonnet 4.5, and Claude Sonnet 4.6) were employed to translate text, correct grammar, and restructure sentences into the required third-person past tense and formal academic style.

No core scientific content, model selection decisions, hyperparameter choices, or interpretation of results were delegated to or generated by the AI model.

7. References

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8. Lists

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